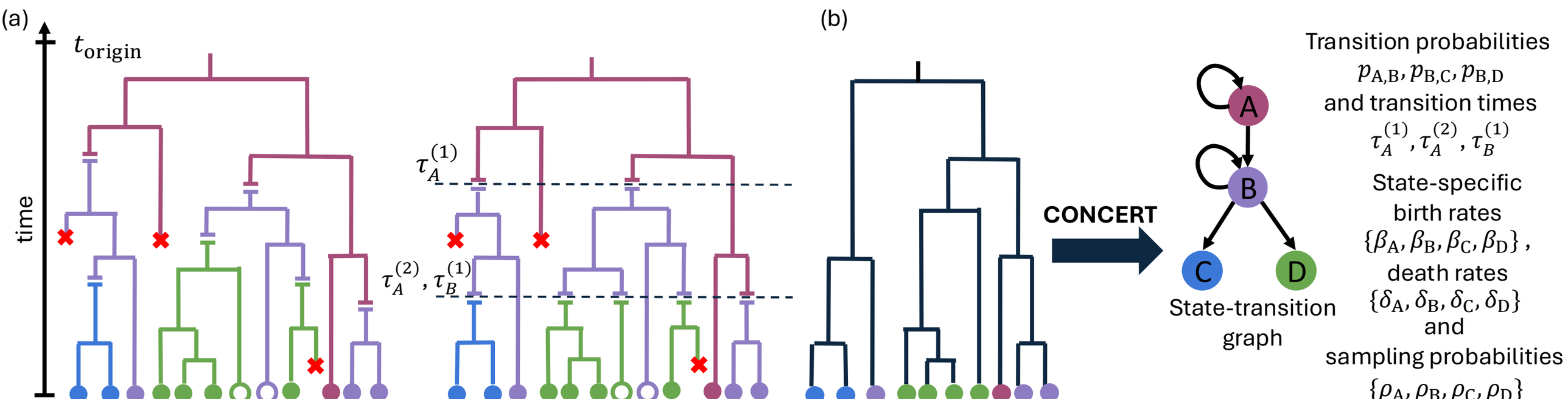

(a)
t_{origin}
time
$\tau_A^{(1)}$
$\tau_A^{(2)}, \tau_B^{(1)}$
(b)
CONCERT
A
B
C
D
State-transition graph
Transition probabilities
$p_{\text{A,B}}, p_{\text{B,C}}, p_{\text{B,D}}$
and transition times
$\tau_A^{(1)}, \tau_A^{(2)}, \tau_B^{(1)}$
State-specific birth rates
$\{\beta_{\text{A}}, \beta_{\text{B}}, \beta_{\text{C}}, \beta_{\text{D}}\}$,
death rates
$\{\delta_{\text{A}}, \delta_{\text{B}}, \delta_{\text{C}}, \delta_{\text{D}}\}$
and
sampling probabilities
$\{\rho_{\text{A}}, \rho_{\text{B}}, \rho_{\text{C}}, \rho_{\text{D}}\}$
Multi-type birth-death model (uncoordinated transitions)
Coordinated state transition model
Observed cell lineage tree
Inferred (identifiable) phylodynamics parameters